Chapter 9. Automorphisms and Galois Theory

Field automorphisms, isomorphism extension, splitting fields, separability, Galois theory, cyclotomic extensions, the quintic

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Throughout, $\overline{F}$ is a fixed algebraic closure of F and all algebraic extensions of F are taken inside it.

9.1 Automorphisms of fields (§48)

Motivation. Galois's idea was to study a polynomial through the symmetries of its roots. Concretely, these are the automorphisms of a field extension that fix the base field; they permute the roots of each polynomial over the base field. This section collects the basic examples, conjugation isomorphisms and the Frobenius map.

α, β algebraic over F are **conjugate over** F if $\operatorname{irr}(\alpha, F) = \operatorname{irr}(\beta, F)$.

Theorem 9.1 (48.3, conjugation isomorphisms)

If α, β are algebraic over F with $\deg(\alpha, F) = n$, the map $\psi_{\alpha,\beta}(c_0 + c_1\alpha + \cdots + c_{n-1}\alpha^{n-1}) = c_0 + c_1\beta + \cdots + c_{n-1}\beta^{n-1}$ is an isomorphism $F(\alpha) \to F(\beta)$ iff α and β are conjugate over F.

Proof. If $\psi_{\alpha,\beta}$ is an isomorphism, it fixes F, so $\operatorname{irr}(\alpha, F)(\beta) = \psi(\operatorname{irr}(\alpha, F)(\alpha)) = 0$, hence $\operatorname{irr}(\beta, F) = \operatorname{irr}(\alpha, F)$. Conversely, if $p = \operatorname{irr}(\alpha, F) = \operatorname{irr}(\beta, F)$, the evaluation maps give isomorphisms $F[x]/\langle p\rangle \to F(\alpha)$ and $F[x]/\langle p\rangle \to F(\beta)$ (Theorem 29.18); $\psi_{\alpha,\beta}$ is the composite of the inverse of the first with the second. ■

Corollary 9.2 (48.5)

An isomorphism of $F(\alpha)$ onto a subfield of $\overline{F}$ fixing F maps α to a conjugate of α; conversely, for each conjugate β there is exactly one such isomorphism with $\alpha \mapsto \beta$, namely $\psi_{\alpha,\beta}$.

Proof. $\psi(\alpha)$ is a zero of $\operatorname{irr}(\alpha, F)$; and an isomorphism fixing F is determined by the image of α. ■

Corollary 9.3 (48.6)

If $f \in \mathbb{R}[x]$ and $f(a + bi) = 0$, then $f(a - bi) = 0$.

Proof. Complex conjugation is $\psi_{i,-i} : \mathbb{R}(i) \to \mathbb{R}(i)$, fixing the coefficients of f. ■

An isomorphism of E onto itself is an **automorphism**. For automorphisms σ_i, the **fixed field** is $E_{\{\sigma_i\}} = \{a : \sigma_i(a) = a\ \forall i\}$.

Theorem 9.4 (48.11)

$E_{\{\sigma_i\}}$ is a subfield of E.

Proof. Each σ_i preserves sums, products, negatives and inverses. ■

Theorem 9.5 (48.14)

The automorphisms of a field form a group under composition.

Theorem 9.6 (48.15)

For $F \leq E$, the automorphisms of E fixing F form a subgroup $G(E/F)$, whose fixed field contains F.

Proof. Composites and inverses of automorphisms fixing F fix F. ■

Theorem 9.7 (48.19, Frobenius)

For a finite field F of characteristic p, $\sigma_p(a) = a^p$ is an automorphism of F with fixed field $\mathbb{Z}_p$.

Proof. σ_p is a ring homomorphism (Lemma 33.9), one-to-one ($a^p = 0$ only for $a = 0$), hence onto the finite set F. The fixed elements are the zeros of $x^p - x$, i.e. the p elements of $\mathbb{Z}_p$ (Fermat). ■

9.2 The isomorphism extension theorem (§49)

Motivation. To count automorphisms we need to know when an isomorphism between small fields extends to larger ones. The isomorphism extension theorem guarantees such extensions exist, and the number of ways of extending (the index $\{E : F\}$) becomes the key invariant of the next sections.

Theorem 9.8 (49.3, isomorphism extension theorem)

Let E be algebraic over F, $\sigma : F \to F'$ an isomorphism, and $\overline{F'}$ an algebraic closure of F'. Then σ extends to an isomorphism τ of E onto a subfield of $\overline{F'}$.

Proof. Consider pairs (L, λ) with $F \leq L \leq E$ and $\lambda : L \to \overline{F'}$ an isomorphism onto its image extending σ, ordered by extension. The union of a chain is again such a pair, so Zorn's lemma gives a maximal (K, λ). If $K \neq E$, take $\alpha \in E - K$ and $p = \operatorname{irr}(\alpha, K)$; the polynomial $\lambda(p)$ is irreducible over $\lambda[K]$ and has a zero $\beta \in \overline{F'}$. As in Theorem 9.1, $K(\alpha) \cong K[x]/\langle p \rangle \cong \lambda[K][x]/\langle \lambda(p) \rangle \cong \lambda[K](\beta)$ gives an extension of λ to $K(\alpha)$, contradicting maximality. So $K = E$. ■

Corollary 9.9 (49.4)

If $E \leq \overline{F}$ is algebraic over F and $\alpha, \beta \in E$ are conjugate over F, then $\psi_{\alpha,\beta}$ extends to an isomorphism of E onto a subfield of $\overline{F}$.

Corollary 9.10 (49.5)

Any two algebraic closures of F are isomorphic by an isomorphism fixing F.

Proof. Extend id_F to $\tau : \overline{F} \to \overline{F}'$. $\tau[\overline{F}]$ is algebraically closed, and $\overline{F}'$ is algebraic over it, so $\overline{F}' = \tau[\overline{F}]$ (Corollary 31.16). ■

The same argument shows: every isomorphism of $\overline{F}$ into itself fixing F is an automorphism; and every isomorphism $\sigma : F \to F'$ extends to an isomorphism $\overline{F} \to \overline{F'}$.

Theorem 9.11 (49.7)

Let E be a finite extension of F, $\sigma : F \to F'$ an isomorphism and $\overline{F'}$ an algebraic closure of F'. The number of extensions of σ to an isomorphism of E onto a subfield of $\overline{F'}$ is finite, independent of σ, F' and $\overline{F'}$, and at most $[E : F]$.

Proof. First let $E = F(\alpha)$, $p = \operatorname{irr}(\alpha, F)$. Extensions of σ correspond bijectively to the zeros of $\sigma(p)$ in $\overline{F'}$ (an extension sends α to such a zero; conversely each zero β gives one, as in the proof of Theorem 9.8). Extending σ to an isomorphism $\overline{F} \to \overline{F'}$ (remark after Corollary 9.10) carries the zeros of p bijectively to those of $\sigma(p)$, so their number m equals the number of distinct zeros of p in $\overline{F}$; $m \le \deg p = [F(\alpha) : F]$. In general, write $E = F(\alpha_1, \dots, \alpha_n)$; each extension of σ to $F(\alpha_1, \dots, \alpha_{k-1})$ has the same number m_k of extensions to $F(\alpha_1, \dots, \alpha_k)$, by the simple case; so the total is $m_1 \cdots m_n \le \prod [F(\alpha_1, \dots, \alpha_k) : F(\alpha_1, \dots, \alpha_{k-1})] = [E : F]$. ■

The number is the **index** $\{E : F\}$.

Corollary 9.12 (49.10)

If $F \le E \le K$ with K finite over F, then $\{K : F\} = \{K : E\}\{E : F\}$.

Proof. Each of the $\{E : F\}$ extensions of id_F to E has exactly $\{K : E\}$ extensions to K (independence in Theorem 9.11). ■

9.3 Splitting fields (§50)

Motivation. For Galois theory to work, an extension must contain all the conjugates of each of its elements, not just some of them; $\mathbb{Q}(\sqrt[3]{2})$ fails this. Splitting fields are exactly the extensions with this property, and for them every embedding into an algebraic closure is an automorphism.

E with $F \le E \le \overline{F}$ is a **splitting field over** F if it is the smallest subfield of $\overline{F}$ containing F and all zeros in $\overline{F}$ of each polynomial of some set $\{f_i\} \subseteq F[x]$.

Theorem 9.13 (50.3)

E ($F \le E \le \overline{F}$) is a splitting field over F iff every automorphism of $\overline{F}$ fixing F maps E onto itself.

Proof. If E is generated by the zeros of the f_i, an automorphism τ fixing F permutes the zeros of each f_i, so $\tau[E] \subseteq E$, and likewise $\tau^{-1}[E] \subseteq E$. Conversely, suppose the condition holds; we show E is the splitting field of $\{\operatorname{irr}(\alpha, F) : \alpha \in E\}$, i.e. every conjugate β of every $\alpha \in E$ lies in E. $\psi_{\alpha,\beta}$ extends to an isomorphism of $\overline{F}$ into itself fixing F (Theorem 9.8), which is an automorphism (remark after Corollary 9.10); it maps E onto E, so $\beta = \tau(\alpha) \in E$. ■

Corollary 9.14 (50.6)

If $E \le \overline{F}$ is a splitting field over F, every irreducible polynomial in $F[x]$ with a zero in E splits in E.

Proof. Its zeros are the conjugates of the given zero, which lie in E by the proof of Theorem 9.13. ■

Corollary 9.15 (50.7)

If $E \leq \overline{F}$ is a splitting field over F, every isomorphism of E onto a subfield of $\overline{F}$ fixing F is an automorphism of E. If E is a finite splitting field, $\{E : F\} = |G(E/F)|$.

Proof. Such an isomorphism extends to an automorphism of $\overline{F}$ (as above), which maps E onto E. ■

Example 9.1

$\mathbb{Q}(\sqrt[3]{2}, \omega)$ ($\omega = e^{2\pi i/3}$) is the splitting field of $x^3 - 2$, of degree 6; $\mathbb{Q}(\sqrt[3]{2})$ is not a splitting field over $\mathbb{Q}$. The splitting field of $x^{p^n} - x$ over $\mathbb{Z}_p$ is $\mathrm{GF}(p^n)$.

9.4 Separable extensions (§51)

Motivation. For the number of automorphisms to equal the degree, irreducible polynomials must not have repeated roots. Separability is this condition; it holds automatically in characteristic 0 and over finite fields, and it yields the primitive element theorem $E = F(\alpha)$.

α is a zero of f of **multiplicity** ν if $(x - \alpha)^\nu$ divides f in $\overline{F}[x]$ but $(x - \alpha)^{\nu+1}$ does not.

Theorem 9.16 (51.2)

All zeros in $\overline{F}$ of an irreducible $f \in F[x]$ have the same multiplicity.

Proof. For zeros α, β, $\psi_{\alpha,\beta}$ extends to an automorphism τ of $\overline{F}$ fixing F. If $f = (x - \alpha)^\nu g$ with $g(\alpha) \neq 0$, applying τ to coefficients, $f = (x - \beta)^\nu \tau(g)$ with $\tau(g)(\beta) = \tau(g(\alpha)) \neq 0$; by unique factorization in $\overline{F}[x]$ the multiplicity of β is ν. ■

Corollary 9.17 (51.3)

An irreducible $f \in F[x]$ factors in $\overline{F}[x]$ as $a \prod_i (x - \alpha_i)^\nu$ with the α_i its distinct zeros.

Theorem 9.18 (51.6)

If E is finite over F, then $\{E : F\}$ divides $[E : F]$.

Proof. For $E = F(\alpha)$: $[F(\alpha) : F] = \deg \mathrm{irr}(\alpha, F) = \nu \cdot m$ with $m = \{F(\alpha) : F\}$ the number of distinct zeros (Corollary 9.17, Theorem 9.11). In general use a chain of simple extensions, Theorem 31.4 and Corollary 9.12. ■

E is a **separable extension** of F if $\{E : F\} = [E : F]$; α is **separable** over F if $F(\alpha)$ is, i.e. if $\mathrm{irr}(\alpha, F)$ has distinct zeros.

Theorem 9.19 (51.9)

For finite $F \leq E \leq K$: K is separable over F iff K is separable over E and E is separable over F.

Proof. $\{K : F\} = \{K : E\}\{E : F\}$ and $[K : F] = [K : E][E : F]$ with $\{K : E\} \leq [K : E]$ and $\{E : F\} \leq [E : F]$; equality of the products holds iff both equalities hold. ■

Corollary 9.20 (51.10)

A finite extension E of F is separable iff every $\alpha \in E$ is separable over F.

Proof. If E is separable, apply Theorem 9.19 to $F \leq F(\alpha) \leq E$. Conversely let $E = F(\alpha_1, \ldots, \alpha_n)$; $\operatorname{irr}(\alpha_i, F(\alpha_1, \ldots, \alpha_{i-1}))$ divides $\operatorname{irr}(\alpha_i, F)$, so has distinct zeros, and each step is separable; apply Theorem 9.19. ■

Lemma 9.21 (51.11)

If $f = x^n + a_{n-1}x^{n-1} + \cdots + a_0 \in \overline{F}[x]$, $f^m \in F[x]$ and $m \cdot 1 \neq 0$ in F, then $f \in F[x]$.

Proof. Suppose $a_{n-1}, \ldots, a_{n-k+1} \in F$. The coefficient of x^{nm-k} in f^m is ma_{n-k} plus a polynomial expression in $a_{n-1}, \ldots, a_{n-k+1}$ with integer coefficients; it lies in F, so $ma_{n-k} \in F$ and $a_{n-k} \in F$. Induct on k. ■

A field is **perfect** if every finite extension is separable.

Theorem 9.22 (51.13)

Every field of characteristic 0 is perfect.

Proof. Let f be monic irreducible over F with distinct zeros α_i of multiplicity ν, and $g = \prod(x - \alpha_i)$. Then $g^\nu = f \in F[x]$ and $\nu \cdot 1 \neq 0$, so $g \in F[x]$ (Lemma 9.21); $g \mid f$ and f irreducible force $\nu = 1$. So every algebraic element is separable; apply Corollary 9.20. ■

Theorem 9.23 (51.14)

Every finite field is perfect.

Proof. If F is finite and α is algebraic over F, then $F(\alpha)$ is finite, of p^n elements, so α is a zero of $x^{p^n} - x$, which has distinct zeros (Lemma 33.8); $\operatorname{irr}(\alpha, F)$ divides it, so has distinct zeros. ■

Theorem 9.24 (51.15, primitive element theorem)

A finite separable extension E of F is simple: $E = F(\gamma)$.

Proof. If F is finite, use Corollary 33.6. Let F be infinite; by induction on the number of generators it suffices to treat $E = F(\alpha, \beta)$. Let $\sigma_1, \ldots, \sigma_n$ be the $n = \{E : F\} = [E : F]$ distinct isomorphisms of E into $\overline{F}$ fixing F. For $i \neq j$, $(\sigma_i\alpha, \sigma_i\beta) \neq (\sigma_j\alpha, \sigma_j\beta)$, so $\sigma_i\alpha + c\sigma_i\beta = \sigma_j\alpha + c\sigma_j\beta$ for at most one $c \in F$. Choose c outside these finitely many values and put $\gamma = \alpha + c\beta$. The σ_i restricted to $F(\gamma)$ are n distinct isomorphisms (they differ on γ), so $[F(\gamma) : F] \geq \{F(\gamma) : F\} \geq n = [E : F]$, and $F(\gamma) = E$. ■

Corollary 9.25 (51.16)

A finite extension of a field of characteristic 0 is simple.

9.5 Totally inseparable extensions (§52)

Motivation. In characteristic p, separability can fail, for instance for $x^p - t$ over $\mathbb{Z}_p(t)$. This section shows that the bad behaviour can be isolated: every algebraic extension is a separable extension

followed by a totally inseparable one.

A finite extension E of F is **totally inseparable** if $\{E : F\} = 1 < [E : F]$; $\alpha \notin F$ is totally inseparable if $F(\alpha)$ is.

Theorem 9.26 (52.3)

For finite $F < E < K$: K is totally inseparable over F iff K is over E and E is over F.

Proof. $\{K : F\} = \{K : E\}\{E : F\}$ is 1 iff both factors are 1. ■

Corollary 9.27 (52.4)

A finite extension $E > F$ is totally inseparable iff every $\alpha \in E - F$ is totally inseparable over F.

Proof. If E is totally inseparable, then $\{E : F\} = \{E : F(\alpha)\}\{F(\alpha) : F\} = 1$ forces $\{F(\alpha) : F\} = 1$, while $[F(\alpha) : F] > 1$. Conversely if $E = F(\alpha_1, \ldots, \alpha_n)$ with each $\mathrm{irr}(\alpha_i, F)$ having one distinct zero, then so does $\mathrm{irr}(\alpha_i, F(\alpha_1, \ldots, \alpha_{i-1}))$, which divides it; so every step has index 1, and $\{E : F\} = 1$. ■

Theorem 9.28 (52.5)

Let F have characteristic $p \neq 0$ and E be finite over F. Then $\alpha \in E - F$ is totally inseparable over F iff $\alpha^{p^t} \in F$ for some $t \geq 1$. Moreover the separable elements of E form a subfield K (the **separable closure** of F in E), and E is totally inseparable over K unless $E = K$.

Proof. If $\alpha^{p^t} = a \in F$, then $x^{p^t} - a = (x - \alpha)^{p^t}$ has the single zero α, so $\mathrm{irr}(\alpha, F)$ does too. Conversely, if $\mathrm{irr}(\alpha, F) = (x - \alpha)^m$, write $m = p^t k$ with $p \nmid k$; then $(x - \alpha)^m = (x^{p^t} - \alpha^{p^t})^k$, and Lemma 9.21 gives $x^{p^t} - \alpha^{p^t} \in F[x]$; $t \geq 1$ since otherwise $k = m$ and $x - \alpha \in F[x]$.

Separable elements form a subfield by Corollary 9.20 applied to $F(\alpha, \beta)$. For $\alpha \in E$, $f = \mathrm{irr}(\alpha, F)$ can be written $g(x^{p^t})$ with g irreducible and $g' \neq 0$ (if $f' = 0$ then $f \in F[x^p]$; iterate), and then g has distinct zeros ($\gcd(g, g') = 1$), so α^{p^t} is separable, i.e. lies in K. By the first part (over K), E is totally inseparable over K. ■

9.6 Galois theory (§53)

Motivation. This is the culmination of the course. For a finite normal (splitting and separable) extension, intermediate fields correspond one-to-one to subgroups of the Galois group, with inclusions reversed and normal subgroups matching normal extensions. Questions about fields become questions about finite groups, which are much easier to compute with.

A finite extension K of F is a **finite normal extension** if it is a separable splitting field over F; then $G(K/F)$ is its **Galois group**, and $|G(K/F)| = \{K : F\} = [K : F]$ (Corollary 9.15).

Theorem 9.29 (53.2)

If K is finite normal over F and $F \leq E \leq K$, then K is finite normal over E, and $G(K/E)$ is the subgroup of $G(K/F)$ of automorphisms fixing E. Two automorphisms $\sigma, \tau \in G(K/F)$ induce the same isomorphism of E into $\overline{F}$ iff they lie in the same left coset of $G(K/E)$.

Proof. K is separable over E (Theorem 9.19), and the splitting field over F of a set of polynomials is also their splitting field over E. $\sigma|_E = \tau|_E$ iff $\sigma^{-1}\tau$ fixes E. ■

Theorem 9.30 (53.6, main theorem of Galois theory)

Let K be finite normal over F, $G = G(K/F)$, and for $F \le E \le K$ let $\lambda(E) = G(K/E)$. Then:
(1) λ is a bijection from the intermediate fields onto the subgroups of G;
(2) $\lambda(E) = G(K/E)$ and $E = K_{G(K/E)}$;
(3) for $H \le G$, $\lambda(K_H) = H$;
(4) $[K : E] = |\lambda(E)|$ and $[E : F] = (G : \lambda(E))$;
(5) E is normal over F iff $\lambda(E) \trianglelefteq G$, and then $G(E/F) \cong G/\lambda(E)$;
(6) λ reverses inclusions.

Proof. (2) Clearly $E \subseteq K_{G(K/E)}$. If $\alpha \in K - E$, $\mathrm{irr}(\alpha, E)$ has degree > 1 and distinct zeros, all in K (Corollary 9.14); for a zero $\beta \neq \alpha$, $\psi_{\alpha,\beta}$ extends to an automorphism of K fixing E (Theorem 9.8 and Corollary 9.15), which moves α. So $E = K_{G(K/E)}$, and λ is one-to-one.

(3) Let $E = K_H$; then $H \le \lambda(E)$. K is separable over E, so $K = E(\alpha)$ (Theorem 9.24). The polynomial $\prod_{\sigma\in H}(x - \sigma\alpha)$ has coefficients fixed by H, i.e. in E, and has α as a zero, so $[K : E] = \deg(\alpha, E) \le |H|$. But $|\lambda(E)| = |G(K/E)| = [K : E]$. Hence $H = \lambda(E)$; this proves surjectivity in (1).

(4) $|\lambda(E)| = [K : E]$, and $[E : F] = [K : F]/[K : E] = |G|/|\lambda(E)|$.

(5) For $\sigma \in G$, $\lambda(\sigma[E]) = \sigma\lambda(E)\sigma^{-1}$. E is a splitting field over F iff every isomorphism of E into $\overline{F}$ fixing F maps E onto E (Theorem 9.13); these isomorphisms are restrictions of elements of G (extend to K, which is a splitting field); so E is normal iff $\sigma[E] = E$ for all $\sigma \in G$, iff $\sigma\lambda(E)\sigma^{-1} = \lambda(E)$ for all σ (by (1)). Then restriction $G \to G(E/F)$ is a homomorphism, onto (every automorphism of E over F extends to K), with kernel $\lambda(E)$.

(6) $E_1 \subseteq E_2$ implies $G(K/E_2) \le G(K/E_1)$, and conversely by (2). ■

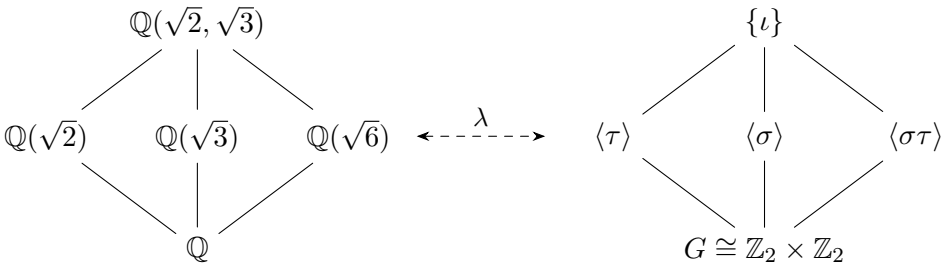


Figure. *The Galois correspondence for $K = \mathbb{Q}(\sqrt{2},\sqrt{3})$, with $\sigma : \sqrt{2} \mapsto -\sqrt{2}$ and $\tau : \sqrt{3} \mapsto -\sqrt{3}$. Each intermediate field is the fixed field of the subgroup in the same position; the lattices are inverted (larger fields correspond to smaller groups).*

Theorem 9.31 (53.7)

If K is an extension of degree n of a finite field F of p^r elements, then $G(K/F)$ is cyclic of order n, generated by $\sigma_{p^r}(\alpha) = \alpha^{p^r}$.

Proof. K is the splitting field of $x^{p^{rn}} - x$ over F, separable (Theorem 9.23), so normal with $|G(K/F)| = n$. σ_{p^r} fixes F (whose elements satisfy $a^{p^r} = a$). If $\sigma_{p^r}^m = \mathrm{id}$ with $0 < m < n$, every $\alpha \in K$ would be a zero of $x^{p^{rm}} - x$, which has fewer than p^{rn} zeros. So σ_{p^r} has order n. ■

Example 9.2

For $K=\mathbb{Q}(\sqrt{2},\sqrt{3})$, $G(K/\mathbb{Q})\cong\mathbb{Z}_2\times\mathbb{Z}_2$, and the three subgroups of order 2 correspond to $\mathbb{Q}(\sqrt{2})$, $\mathbb{Q}(\sqrt{3})$, $\mathbb{Q}(\sqrt{6})$. For $K=\mathbb{Q}(\sqrt[3]{2},\omega)$, $G\cong S_3$; the normal subgroup A_3 corresponds to the normal extension $\mathbb{Q}(\omega)$.

Algebraic proof of the fundamental theorem of algebra. Let K be a finite normal extension of $\mathbb{R}$ containing i, with $G=G(K/\mathbb{R})$ of order $2^k m$, m odd. The fixed field of a Sylow 2-subgroup has odd degree m over $\mathbb{R}$; by the intermediate value theorem every real polynomial of odd degree has a real zero, so $m=1$. Then $G(K/\mathbb{C})$ is a 2-group; if nontrivial it has a subgroup of index 2, giving a quadratic extension of $\mathbb{C}$, impossible since every complex number has a square root. So $K=\mathbb{C}$.

9.7 Illustrations of Galois theory (§54)

Motivation. The correspondence is best understood through examples: computing the Galois group of a splitting field and drawing the subgroup and subfield lattices side by side. Symmetric functions also appear here as the fixed field of the full symmetric group.

For indeterminates $y_1,\dots,y_n$ the **elementary symmetric functions** s_k are defined by $\prod(x-y_i)=x^n-s_1x^{n-1}+s_2x^{n-2}-\cdots+(-1)^n s_n$.

Theorem 9.32 (54.2)

Every symmetric function of $y_1,\dots,y_n$ over F is a rational function of $s_1,\dots,s_n$: $F(y_1,\dots,y_n)^{S_n}=F(s_1,\dots,s_n)$.

Proof. Let $K=F(y_1,\dots,y_n)$ and $E=F(s_1,\dots,s_n)\le K$. K is the splitting field over E of $\prod(x-y_i)\in E[x]$, which has distinct zeros, so K is finite normal over E with $[K:E]\le n!$ (adjoin the zeros one at a time). S_n acts on K by permuting the y_i, by automorphisms fixing E; so $|G(K/E)|\ge n!$, hence $=n!$, and $G(K/E)=S_n$. By Theorem 9.30(2), $E=K_{S_n}$, the field of symmetric functions. ∎

Example 9.3

$y_1^2+\cdots+y_n^2=s_1^2-2s_2$. The Galois group of x^4-2 over $\mathbb{Q}$ is the dihedral group D_4 of order 8, and its lattice of 10 subgroups matches the lattice of intermediate fields of $\mathbb{Q}(\sqrt[4]{2},i)$.

9.8 Cyclotomic extensions (§55)

Motivation. Adjoining roots of unity produces abelian Galois groups, and these extensions are the first step of every radical tower. As an application we obtain Gauss's characterization of the regular polygons constructible with ruler and compass.

The splitting field of x^n-1 over F is the **n-th cyclotomic extension**. Over $\mathbb{Q}$ it is $\mathbb{Q}(\zeta)$ with $\zeta=e^{2\pi i/n}$; the primitive n-th roots of unity are the ζ^k with $\gcd(k,n)=1$, zeros of the **cyclotomic polynomial** $\Phi_n(x)=\prod_{\gcd(k,n)=1}(x-\zeta^k)$, which lies in $\mathbb{Z}[x]$ (from $x^n-1=\prod_{d\mid n}\Phi_d$ by induction).

Theorem 9.33 (55.4)

The Galois group of the n-th cyclotomic extension of $\mathbb{Q}$ has $\phi(n)$ elements and is isomorphic to the group G_n of units of $\mathbb{Z}_n$.

Proof.

Step 1. [Supplement] (Φ_n is irreducible over $\mathbb{Q}$.) Let $f = \text{irr}(\zeta, \mathbb{Q})$; by Gauss's lemma $f \in \mathbb{Z}[x]$ and $x^n - 1 = fg$ with $g \in \mathbb{Z}[x]$ monic. Let p be a prime not dividing n; we show $f(\zeta^p) = 0$. Otherwise $g(\zeta^p) = 0$, so ζ is a zero of $g(x^p)$ and $f(x) \mid g(x^p)$ in $\mathbb{Z}[x]$. Reduce modulo p: $\bar{g}(x^p) = \bar{g}(x)^p$ in $\mathbb{Z}_p[x]$, so $\bar{f} \mid \bar{g}^p$ and $\bar{f}, \bar{g}$ have a common zero in $\overline{\mathbb{Z}_p}$. Then $x^n - \bar{1} = \bar{f}\bar{g}$ has a repeated zero; but its derivative nx^{n-1} is nonzero at every zero ($p \nmid n$), a contradiction. Applying this repeatedly to the prime factors of k, every ζ^k with $\gcd(k, n) = 1$ is a zero of f; hence $f = \Phi_n$ and $[\mathbb{Q}(\zeta) : \mathbb{Q}] = \phi(n)$.

Step 2. An automorphism σ is determined by $\sigma(\zeta)$, a primitive root ζ^k with $k \in G_n$; $\sigma \mapsto k$ is a homomorphism ($\sigma\tau(\zeta) = \sigma(\zeta^l) = \zeta^{kl}$), one-to-one, and onto since both groups have $\phi(n)$ elements. ■

Corollary 9.34 (55.6)

For a prime p, the Galois group of the p-th cyclotomic extension of $\mathbb{Q}$ is cyclic of order $p - 1$.

Proof. $G_p = \mathbb{Z}_p^*$ is cyclic (Corollary 23.6). ■

Theorem 9.35 (55.8, Gauss)

The regular n-gon is constructible with straightedge and compass iff all odd primes dividing n are Fermat primes ($2^{2^k} + 1$) whose squares do not divide n.

Proof. The n-gon is constructible iff $\cos(2\pi/n)$ is constructible. Since ζ is a zero of $x^2 - 2\cos(2\pi/n)x + 1$, $[\mathbb{Q}(\zeta) : \mathbb{Q}(\cos 2\pi/n)] = 2$ for $n \geq 3$. If $\cos(2\pi/n)$ is constructible, its degree is a power of 2 (Corollary 32.8), so $\phi(n)$ is a power of 2. Conversely, if $\phi(n) = 2^m$, the Galois group G_n is an abelian 2-group; the subgroup fixing $\mathbb{Q}(\cos 2\pi/n)$ has order 2, and the quotient, an abelian group of order 2^{m-1}, has a chain of subgroups each of index 2 (Theorem 36.8). By the Galois correspondence this gives a tower $\mathbb{Q} = L_0 < L_1 < \cdots < L_{m-1} = \mathbb{Q}(\cos 2\pi/n)$ of real fields with $[L_{j+1} : L_j] = 2$; each $L_{j+1} = L_j(\sqrt{d_j})$ with $0 < d_j \in L_j$, so $\cos(2\pi/n)$ is constructible (Theorem 32.6). Finally $\phi(n) = \prod p^{a-1}(p-1)$ is a power of 2 iff every odd $p \mid n$ has $a = 1$ and $p - 1$ a power of 2; and $2^s + 1$ prime forces s to be a power of 2. ■

So the regular 17-gon is constructible, and the regular 7-gon and 9-gon are not.

9.9 Insolvability of the quintic (§56)

Motivation. We can now answer the question that motivated Galois. A polynomial is solvable by radicals exactly when its Galois group is solvable; since S_5 is not solvable (A_5 is simple), there are quintic polynomials over $\mathbb{Q}$ whose roots cannot be written using $+, -, \times, \div$ and radicals.

An extension K of F is an **extension by radicals** if $K = F(\alpha_1, \ldots, \alpha_r)$ with $\alpha_i^{n_i} \in F(\alpha_1, \ldots, \alpha_{i-1})$; $f \in F[x]$ is **solvable by radicals** if its splitting field lies in such a K. Fields have characteristic 0 in this section.

Lemma 9.36 (56.3)

If K is the splitting field of $x^n - a$ over F, then $G(K/F)$ is solvable.

Proof. If $a = 0$ there is nothing to show. The n zeros of $x^n - a$ are distinct (the derivative nx^{n-1} does not vanish at them); if α is one, the quotients α'/α are n distinct n-th roots of unity in K, forming a cyclic group (Corollary 23.6) with a generator ζ, a primitive n-th root of unity. So $K = F(\zeta, \alpha)$. $F(\zeta)$ is normal over F with $G(F(\zeta)/F)$ isomorphic to a subgroup of G_n, abelian. An element of $G(K/F(\zeta))$ is determined by $\alpha \mapsto \zeta^k \alpha$, and $\sigma \mapsto k$ is a one-to-one homomorphism into $\mathbb{Z}_n$; so $G(K/F(\zeta))$ is abelian. The series $\{e\} \leq G(K/F(\zeta)) \trianglelefteq G(K/F)$ has abelian factors (Theorem 9.30(5)). ■

Theorem 9.37 (56.4)

If $F \leq E \leq K$ with E normal over F and K an extension of F by radicals, then $G(E/F)$ is solvable.

Proof. Let $K = F(\alpha_1, \ldots, \alpha_r)$ as above, $n = \prod n_i$, and L the splitting field over F of $(x^n - 1)\prod_i \operatorname{irr}(\alpha_i, F)$; $L \supseteq K \supseteq E$ and L is normal over F. L is generated over $F(\zeta)$ (ζ a primitive n-th root of unity) by the images $\sigma(\alpha_i)$, $\sigma \in G(L/F)$, and $\sigma(\alpha_i)^{n_i} \in F(\sigma(\alpha_1), \ldots, \sigma(\alpha_{i-1}))$. Ordering these generators accordingly gives a tower $F \leq F(\zeta) = L_0 \leq L_1 \leq \cdots \leq L_s = L$ with $L_j = L_{j-1}(\beta_j)$, $\beta_j^{m_j} \in L_{j-1}$, $m_j \mid n$. Since L_{j-1} contains the m_j-th roots of unity, L_j is the splitting field of $x^{m_j} - \beta_j^{m_j}$ over L_{j-1}, normal with abelian Galois group (proof of Lemma 9.36). Then $G(L/L_s) \leq G(L/L_{s-1}) \leq \cdots \leq G(L/L_0) \leq G(L/F)$ is a subnormal series (Theorem 9.30(5)) whose factors $G(L_j/L_{j-1})$ and $G(F(\zeta)/F)$ are abelian. So $G(L/F)$ is solvable, and so is its factor group $G(E/F) \cong G(L/F)/G(L/E)$ (factor groups of solvable groups are solvable, by Theorem 34.5 applied to a series). ■

Theorem 9.38 (56.6)

If $y_1, \ldots, y_5$ are independent transcendentals over $\mathbb{Q}$, then $f(x) = \prod_{i=1}^{5}(x - y_i)$ is not solvable by radicals over $F = \mathbb{Q}(s_1, \ldots, s_5)$.

Proof. By the proof of Theorem 9.32, the splitting field $\mathbb{Q}(y_1, \ldots, y_5)$ has Galois group S_5 over F. S_5 is not solvable: its composition series $\{e\} < A_5 < S_5$ has the nonabelian simple factor A_5 (Theorem 15.15, Jordan–Hölder). By Theorem 9.37, f is not solvable by radicals. ■

Consequently there is no general formula by radicals for the zeros of a polynomial of degree 5; concretely, $x^5 - 4x + 2$ (irreducible by Eisenstein, with exactly three real zeros, so its Galois group contains a 5-cycle and a transposition and is S_5) is not solvable by radicals over $\mathbb{Q}$.